LAB 01-VIRTUAL ENVIRONMENT PREPARATION

JUNE 2, 2026

NET46
CHANG LIU

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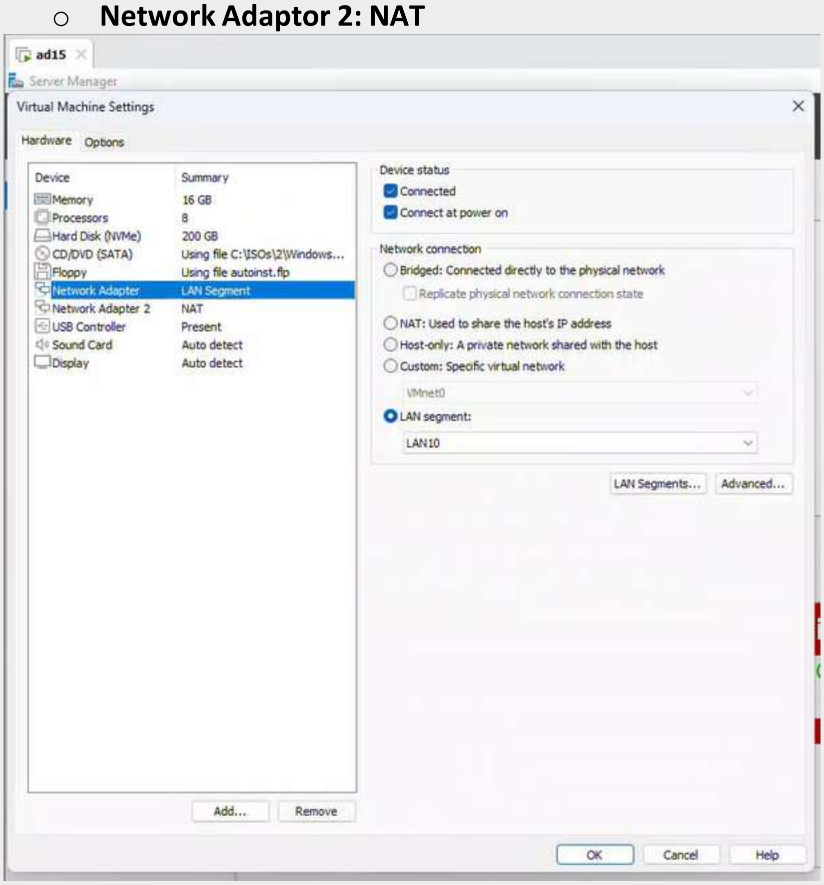
Exercise 1 – Installing Windows Server 2022

Objective:

- Create a **Windows server 2022 VM.**

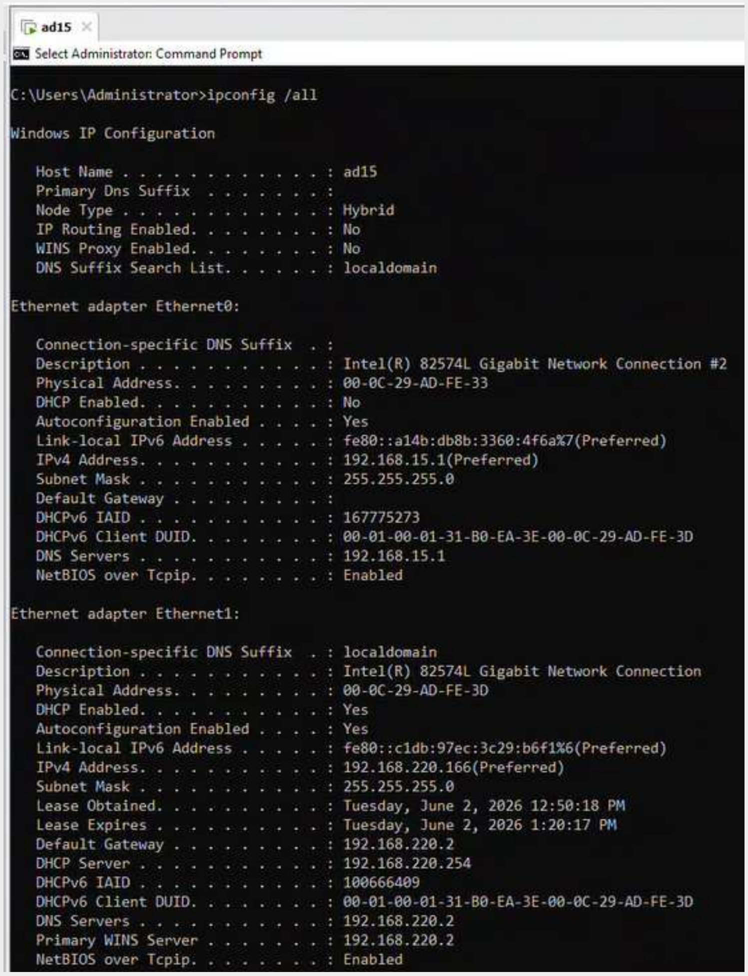
Tasks:

1. **Create a VM with the following specifications:**
    - OS: Windows Server 2022 Datacenter Edition
    - Serial number: **7FW3K-3NWHH-R8WFG-GDTRR-R3PY7**
    - Use **administrator** as username with the password **Passw0rd$**
    - Name the VM: **adXX** – *Where XX is your remote computer number, ex **ad01***
    - Specs: 8 vCPU, 16 GB RAM, 200 GB HDD
    - Network interfaces:
        - Network Adaptor: LAN Segment: LAN10
        - **Network Adaptor 2: NAT**



2. **After the installation:**

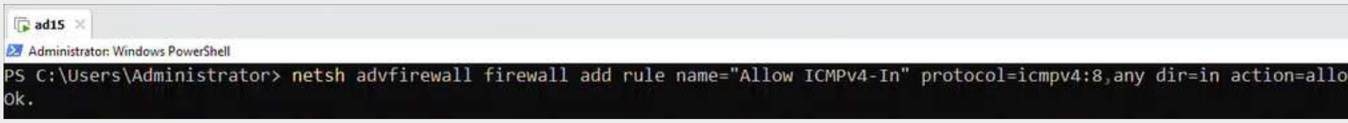
- VMware Tools will be installed automatically; you will need to restart the machine.
- Assign for LAN10 a static IP: 192.168.X.1/24 (Where X is your remote computer number).
- Set Primary DNS Server: 192.168.X.1
- Name the server adXX – Where XX is your remote computer number, ex ad01.



```
ad15
Select Administrator: Command Prompt

C:\Users\Administrator>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : ad15
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No
   DNS Suffix Search List. . . . . . : localdomain

Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection #2
   Physical Address. . . . . . . . . : 00-0C-29-AD-FE-33
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::a14b:db8b:3360:4f6a%7(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.15.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 167775273
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-31-B0-EA-3E-00-0C-29-AD-FE-3D
   DNS Servers . . . . . . . . . . . : 192.168.15.1
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Ethernet1:

   Connection-specific DNS Suffix  . : localdomain
   Description . . . . . . . . . . . : Intel(R) 82574L Gigabit Network Connection
   Physical Address. . . . . . . . . : 00-0C-29-AD-FE-3D
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::c1db:97ec:3c29:b6f1%6(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.220.166(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Tuesday, June 2, 2026 12:50:18 PM
   Lease Expires . . . . . . . . . . : Tuesday, June 2, 2026 1:20:17 PM
   Default Gateway . . . . . . . . . : 192.168.220.2
   DHCP Server . . . . . . . . . . . : 192.168.220.254
   DHCPv6 IAID . . . . . . . . . . . : 100666409
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-31-B0-EA-3E-00-0C-29-AD-FE-3D
   DNS Servers . . . . . . . . . . . : 192.168.220.2
   Primary WINS Server . . . . . . . : 192.168.220.2
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

- Enable ping using the following command:

  netsh advfirewall firewall add rule name="Allow ICMPv4-In" protocol=icmpv4:8,any dir=in action=allow



```
ad15
Administrator: Windows PowerShell
PS C:\Users\Administrator> netsh advfirewall firewall add rule name="Allow ICMPv4-In" protocol=icmpv4:8,any dir=in action=allo
Ok.
```

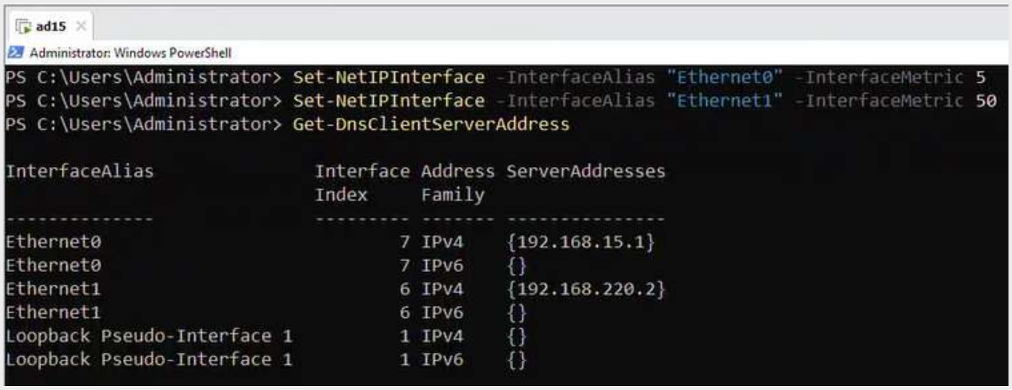
- Configure DNS Interface Metrics:

  Ensure the **LAN NIC (Ethernet0)** is preferred by assigning it a lower metric, and set a higher metric for the **NAT NIC (Ethernet1):**

  Set-NetIPInterface -InterfaceAlias "Ethernet0" -InterfaceMetric 5

Set-NetIPInterface -InterfaceAlias "Ethernet1" -InterfaceMetric 50

Verify DNS Server Configuration
Get-DnsClientServerAddress



- Restart the machine.

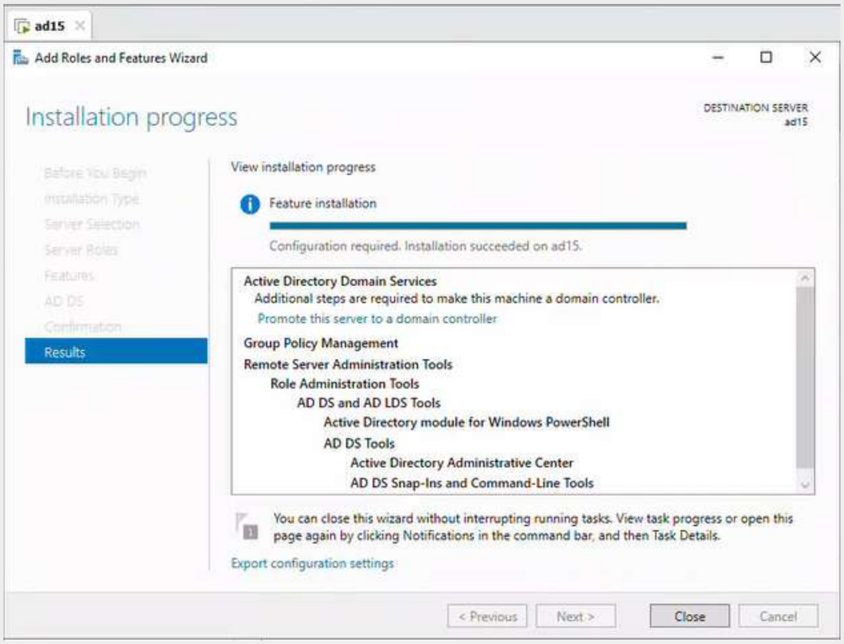
Exercise 2 – Installing the Active Directory role

Objective:

- **Install an Active Directory domain controller.**

Tasks:

1. Open the Server Manager console.
2. In the Select Installation Type window, keep the default Role- or Feature-based Installation option, and then click the Next button.
3. The Select Destination Server window appears. In this window, keep the default option Select a server from the server pool and then click the Next button: the Select Server Roles window appears.
4. Select Active Directory Domain Services, another window appears, click Add Features then click the Next button.
5. In the Select Features window, keep the default options and click the Next button.
6. In the Active Directory Domain Services window, you find information about the need to have a DNS installed on the network, click the Next button.
7. The Confirmation window appears. In this window, check the list of features that will be installed.
8. Do not select the box Automatically restart the destination server, it is not necessary
9. If everything seems correct, click on the Install button.



10. Once the installation is complete, click on the Promote this server as a domain controller link.

11. Select Add New Forest and name it domainXX.local domain (where XX is your number), then click Next.

12. Keep the Forest Functional Level to Windows 2016 and enter the password: Passw0rd$ then click Next.

13. Do not check Create DNS Delegation as it will be the first DNS, then click Next.

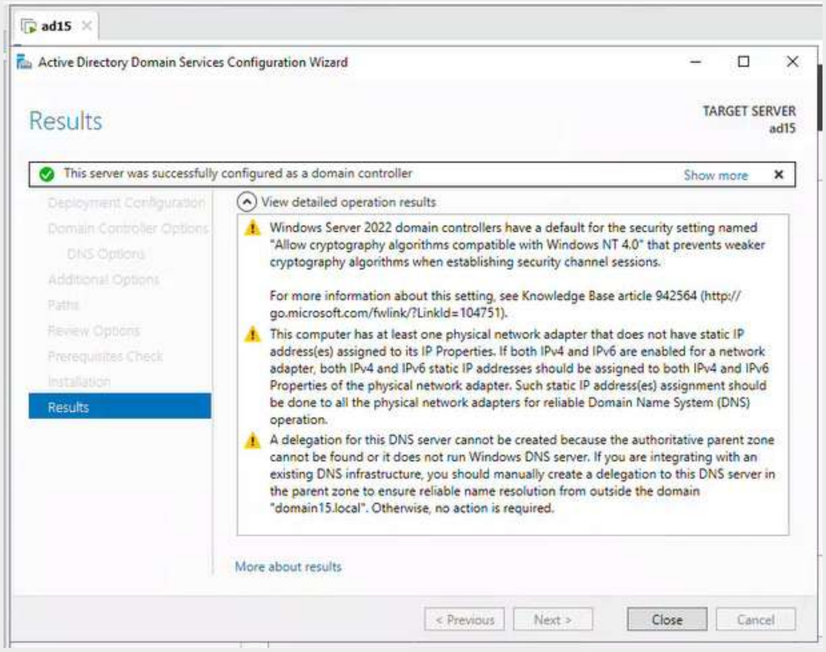
14. Keep the NetBIOS domain name selected, then click Next.

15. Keep the default options for database locations, log files, and SYSVOL, and then click Next.

16. Check the summary of selections and then click Next.

17. In the Verify System Requirements window, click Install. (Ignore the Warnings).

18. The server will restart to complete the configuration.



Exercise 3 – Installing the IIS 10

<u>Objective:</u>

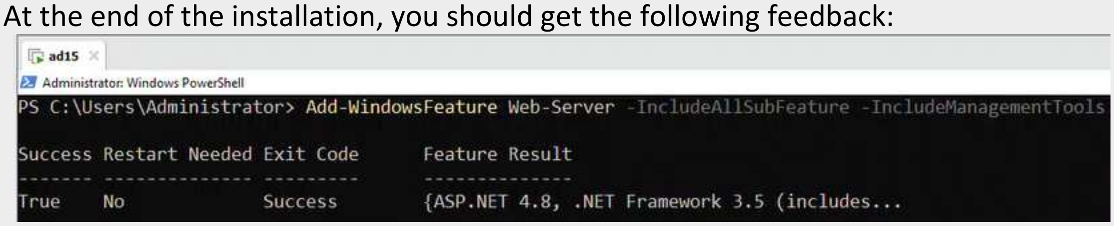
- Install IIS 10.

Tasks

1. Open Windows PowerShell.
2. You are going to install all the features of IIS using PowerShell,
3. In the Windows PowerShell console, type the following command on a single line:

   Add-WindowsFeature Web-Server -IncludeAllSubFeature -IncludeManagementTools

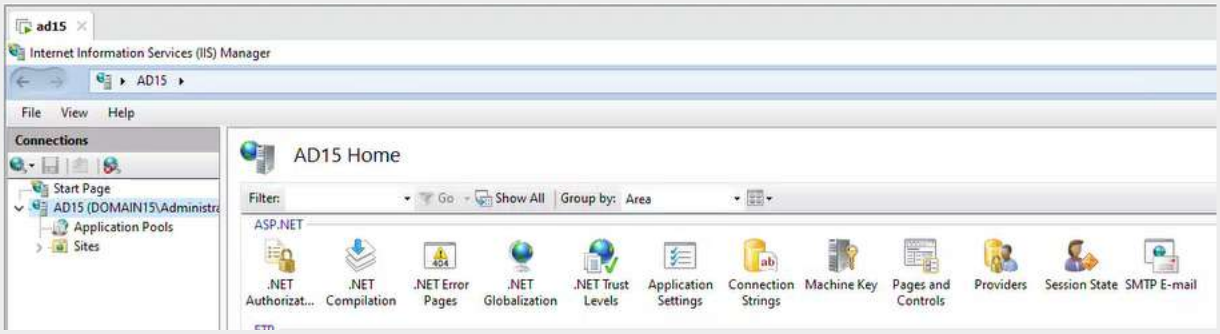
   - The -IncludeAllSubFeature parameter indicates that all web server functionality must be installed.
   - The -IncludeManagementTools parameter indicates that management tools, such as Internet Information Services (IIS) Manager, should also be installed.
4. Press Enter: a message is displayed in the form of a banner, and it warns you that the installation is starting.
5. Wait for the installation time which may take several minutes.
6. At the end of the installation, you should get the following feedback:



7. If the return code Exit Code is at Success, it means that the installation went well.

3. Verify that the installation was also successful by opening **Server Manager**, click on the **Tools** menu, and select **Internet Information Services (IIS) Manager**.

4. Make sure you have this window.



8. Close all windows and shutdown the VM.
9. Create a copy of this VM using Export to adXX.ova

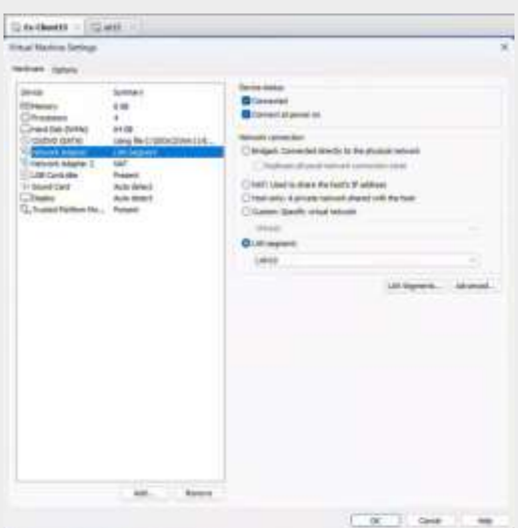
Exercise 4 – Create Windows 11 Client

Objective:

- Create a **Windows 11 VM.**

Tasks:

1. Create a VM with the following specifications:
   - OS: Windows Server 11 Education
   - Serial number: 237NF-QT8QD-YHC2M-D8CG2-76DFM
   - Name the VM Ex-ClientXX – Where XX is your remote computer number.
   - Specs: 4 vCPU, 6 GB RAM, 64 GB HDD
   - Network interfaces:
     - **Network Adaptor: LAN Segment: LAN10**
     - **Network Adaptor 2: NAT**



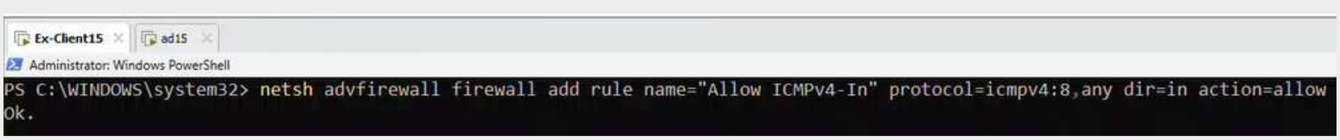
- During the setup at this step - **Create a local user not Microsoft**
    - Click on **shift+f10**
    - Type: **start ms-cxh:localonly**
    - Create a user **using your name** with password: **Passw0rd$**
    - Answer test to all 3 random questions

2. After the installation:

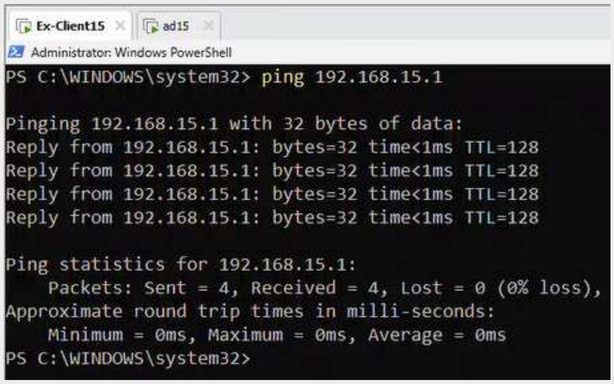
- **VMware Tools** will be installed automatically; you will need to **restart** the machine.
- Assign for **LAN10** a static IP: **192.168.*X*.100/24 (***Where X is your remote computer number).*
- Set Primary DNS Server: **192.168.*X*.1**

- Enable ping using the following command:

  netsh advfirewall firewall add rule name="Allow ICMPv4-In" protocol=icmpv4:8,any dir=in action=allow



- Try to ping the server: **192.168.*X*.1**



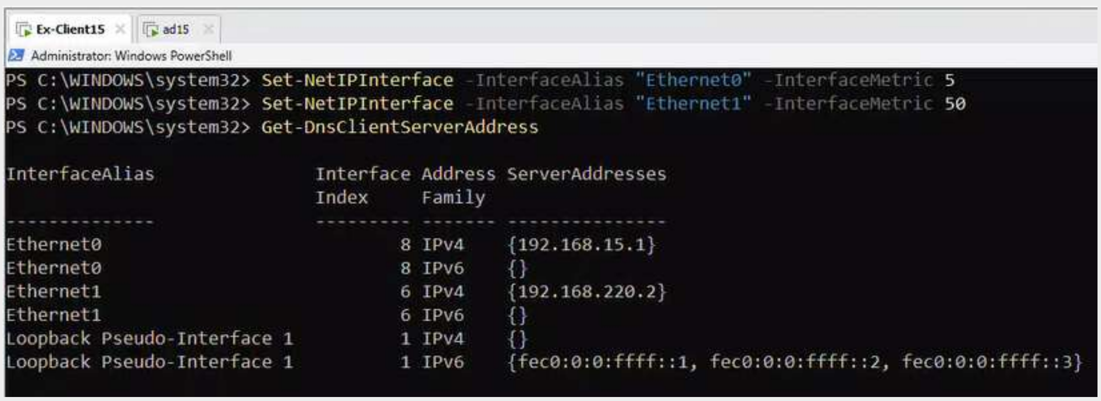
```
Ex-Client15   ad15
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> ping 192.168.15.1

Pinging 192.168.15.1 with 32 bytes of data:
Reply from 192.168.15.1: bytes=32 time<1ms TTL=128
Reply from 192.168.15.1: bytes=32 time<1ms TTL=128
Reply from 192.168.15.1: bytes=32 time<1ms TTL=128
Reply from 192.168.15.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.15.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\WINDOWS\system32>
```

- Configure DNS Interface Metrics: **(Run PowerShell as Administrator)**

  Ensure the **LAN NIC (Ethernet0)** is preferred by assigning it a lower metric, and set a higher metric for the **NAT NIC (Ethernet1):**

  Set-NetIPInterface -InterfaceAlias "Ethernet0" -InterfaceMetric 5

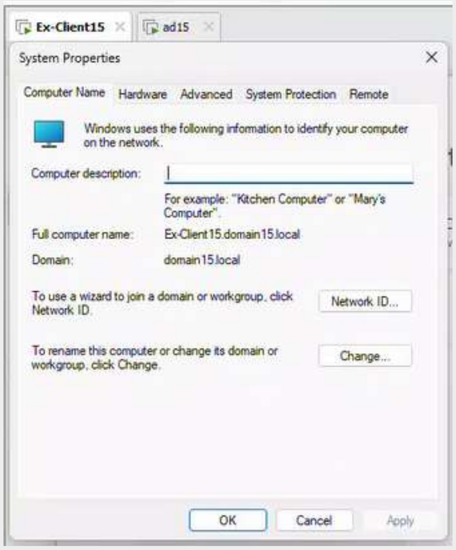
  Set-NetIPInterface -InterfaceAlias "Ethernet1" -InterfaceMetric 50 Verify DNS Server Configuration
  Get-DnsClientServerAddress



```
Ex-Client15   ad15
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> Set-NetIPInterface -InterfaceAlias "Ethernet0" -InterfaceMetric 5
PS C:\WINDOWS\system32> Set-NetIPInterface -InterfaceAlias "Ethernet1" -InterfaceMetric 50
PS C:\WINDOWS\system32> Get-DnsClientServerAddress

InterfaceAlias               Interface Address ServerAddresses
                             Index     Family
--------------               --------- ------- ---------------
Ethernet0                            8 IPv4    {192.168.15.1}
Ethernet0                            8 IPv6    {}
Ethernet1                            6 IPv4    {192.168.220.2}
Ethernet1                            6 IPv6    {}
Loopback Pseudo-Interface 1          1 IPv4    {}
Loopback Pseudo-Interface 1          1 IPv6    {fec0:0:0:ffff::1, fec0:0:0:ffff::2, fec0:0:0:ffff::3}
```

- Name the computer: **Ex-Client*XX*** *(Where **XX** is your remote computer number).*
- Restart the machine.

3. Join Domain:
   - **Join this computer to your domain** then login with the **domainXX\administrator** to test it.



4. Install Microsoft Office:

- Use the following link, download **OfficeSetup.exe**:

  [Download Microsoft Office]

- Once downloaded, double-click on **OfficeSetup.exe** to install **Microsoft office.**